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LAUNDRY MANAGEMENT SYSTEM

XAMPP PROJECT

INTRODUCTION:

XAMPP is a free and open-source software package used to create and test web applications on a local computer. It provides all the necessary components required to develop dynamic websites using technologies such as PHP and MySQL. The name XAMPP represents Cross-Platform, Apache, MariaDB/MySQL, PHP, and Perl. XAMPP creates a local server environment in which PHP programs can be developed and executed without uploading them to an online server. The Apache web server processes PHP files, while MariaDB/MySQL is used to store and manage application data. phpMyAdmin provides a graphical interface for creating databases and tables.

COMPONENTS:

HTML –Forcreating structure and content of web pages.
CSS –Fordesigning layout, forms, navigation, and overall appearance.
PHP –Forserver-side processing, connecting website with database.
MySQL –Forstoring destinations, local guide details, and user trip plans.
XAMPP –Provides the Apache web server and MySQL database.
phpMyAdmin –Used to create, import, and manage the project database.
Web Browser –Used to run and test the website through localhost.

FEATURES:

PHP Support– It allows PHP programs to be developed and tested locally.
Database Support – MariaDB/MySQL can be used to create and manage databases.
phpMyAdmin Integration – Databases can be managed through a graphical interface.
Local Testing – Websites can be tested without an internet connection or live server.
Easy Server Management – Apache and database services can be started or stopped using the XAMPP Control Panel.
Suitable for Beginners – Its simple interface makes it useful for students who are learning web development..

HOW XAMPP WORKS:

XAMPP works by creating a local server environment on the user's computer. The basic working process is

User



Web Browser



Apache Web Server



PHP Interpreter



Database (MariaDB/MySQL)



Response



Web Browser

When a user enters a URL such as:

`http://localhost/laundry_system/`

the browser sends a request to the Apache server running on the local computer.

Apache locates the requested PHP file in the XAMPP htdocs folder. The PHP code is then processed by the PHP interpreter. If the program needs database information, PHP communicates with MariaDB/MySQL. The result is then returned to Apache, which sends the generated webpage back to the browser. Thus, XAMPP allows the complete interaction between browser, web server, PHP, and database to take place locally.

STEPS TO USE XAMPP:

The following steps can be followed to create and run a PHP project using XAMPP.

Step 1: Install XAMPP

Download and install XAMPP on the computer.

Step 2: Open XAMPP Control Panel

Launch the XAMPP Control Panel after installation.

Step 3: Start Apache

Click **Start** next to Apache. Apache is required to run PHP files.

Step 4: Start MySQL/MariaDB

Click **Start** next to MySQL/MariaDB when the project requires a database.

Step 5: Create a Project Folder

Create a project folder inside the htdocs directory.

Example: C:\xampp\htdocs\laundry_system

Step 6: Create PHP Files

Create files such as:

index.php , dashboard.php , booking.php , customers.php , history.php , db.php

and place them inside the project folder.

Step 7: Create a Database

Open: <http://localhost/phpmyadmin>

Create the required database and tables.

Step 8: Connect PHP with the Database

PHP can connect to the MariaDB/MySQL database using suitable database connection commands.

Step 9: Run the Project

Open a web browser and enter: http://localhost/laundry_system/

The PHP application will then be executed through the local Apache server.

APPLICATION:

NAME : LAUNDRY MANAGEMENT SYSTEM

ABOUT THE APPLICATION:

The Laundry Management System is a web-based application developed to help laundry shop owners and staff manage their daily operations efficiently. The system helps track customer details, machine status, service types, orders, and billing in a centralized and organized manner.

The main purpose of the project is to digitize traditional laundry shop operations, reducing the need for paper-based records such as notebooks, handwritten slips, and manual billing. Users can visit the website to view machine availability, book washing machines, track order status, manage customers, and generate bills automatically.

The project is developed using HTML and CSS for designing and styling the web pages and PHP for implementing the server-side functionality. XAMPP is used as the local development environment, with Apache acting as the web server and MySQL used for storing and managing project data. PHP allows the website to process user requests and display dynamic information.

The Laundry Management System aims to make laundry operations simple, organized, and efficient for shop owners. By combining customer management, machine tracking, order booking, and billing features into a single application, the system provides a convenient platform for managing daily laundry activities.

SOURCE CODE:

Index.html

```
<!DOCTYPE      html>

<html>

<head>
    <title>Laundry Management System</title>
    <link rel="stylesheet" type="text/css" href="style.css">
</head>

<frameset rows="15%,85%" border="0">
    <frame src="header.php" name="header" scrolling="no">
    <frameset cols="20%,80%" border="1">
        <frame src="menu.php" name="menu" scrolling="no">
        <frame src="dashboard.php" name="main">
    </frameset>
</frameset>

</html>
```

menu.php

```
<!DOCTYPE  html>

<html>

<head>
    <link rel="stylesheet" type="text/css" href="style.css">
</head>

<body>

<div class="menu">
    <h3>📋 Menu</h3>
```

```
<ul>
  <li><a href="dashboard.php" target="main">🏠 Dashboard</a></li>
  <li><a href="machines.php" target="main">🔄 Machine Status</a></li>
  <li><a href="booking.php" target="main">📅 Book Machine</a></li>
  <li><a href="customers.php" target="main">👤 Customers</a></li>
  <li><a href="history.php" target="main">📜 History</a></li>
```

```
</ul>
```

```
</div>
```

```
</body>
```

```
</html>
```

header.php

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <link rel="stylesheet" type="text/css" href="style.css">
```

```
</head>
```

```
<body>
```

```
<div class="header">
```

```
  <h1>🔄 Laundry Management System</h1>
```

```
  <p>Washing Machine Status Dashboard</p>
```

```
</div>
```

```
</body>
```

```
</html>
```

history.php

```
<?php
```

```
include 'db.php';
```

?>

<!DOCTYPE html>

<html>

<head>

<link rel="stylesheet" type="text/css" href="style.css">

</head>

<body>

<div class="history">

<h2>📅 Booking History</h2>

<table border="1" cellpadding="10" cellspacing="0">

<tr>

<th>Booking ID</th>

<th>Customer</th>

<th>Machine</th>

<th>Mode</th>

<th>Cloth Type</th>

<th>Quantity</th>

<th>Total Cost</th>

<th>Status</th>

<th>Date</th>

<th>Action</th>

</tr>

<?php

\$sql = "SELECT b.*, c.name as customer_name, m.machine_name,
s.service_name

FROM bookings b

```

JOIN customers c ON b.customer_id = c.customer_id
JOIN machines m ON b.machine_id = m.machine_id
JOIN services s ON b.service_id = s.service_id
ORDER BY b.booking_id DESC";

```

```

$result = $conn->query($sql);
while($row = $result->fetch_assoc()) {
    echo "<tr>";
    echo "<td>#" . $row['booking_id'] . "</td>";
    echo "<td>" . $row['customer_name'] . "</td>";
    echo "<td>" . $row['machine_name'] . "</td>";
    echo "<td>" . $row['service_name'] . "</td>";
    echo "<td>" . $row['cloth_type'] . "</td>";
    echo "<td>" . $row['quantity'] . "</td>";
    echo "<td>₹" . $row['total_cost'] . "</td>";
    $status_class = $row['status'];
    echo "<td class='$status_class'>" . strtoupper($row['status']) . "</td>";
    echo "<td>" . $row['booking_date'] . "</td>";
    //Status Update Action
    echo "<td>";
    if($row['status'] == 'pending') {
        echo "<a href='update_status.php?id=" . $row['booking_id'] .
"&status=processing' class='status-btn processing-btn'>▶ Start Processing</a>";
    } elseif($row['status'] == 'processing') {
        echo "<a href='update_status.php?id=" . $row['booking_id'] .
"&status=completed' class='status-btn complete-btn'>✅ Mark
Completed</a>";
    }
}

```

```

    } elseif($row['status'] == 'completed') {
        echo "<span class='completed-text'>

```

```

$update_sql = "UPDATE bookings SET status = '$new_status' WHERE
booking_id = $booking_id";

if ($conn->query($update_sql) === TRUE) {
    // If status is completed, free the machine
    if($new_status == 'completed') {
        $free_machine = "UPDATE machines SET status = 'free', current_mode
= NULL, time_remaining = 0 WHERE machine_id = $machine_id";
        $conn->query($free_machine);
    }
    echo "<script>
        alert('✅ Status updated to " . strtoupper($new_status) . " successfully!');
        window.location.href = 'history.php';
    </script>";
} else {
    echo "Error updating status: " . $conn->error;
}
} else {
    header("Location: history.php");
}
$conn->close();
?>

```

machines.php

```

<?php
include 'db.php';
?>
<!DOCTYPE html>
<html>

```

```

<head>

    <link rel="stylesheet" type="text/css" href="style.css">

</head>

<body>

<div class="machines">

    <h2>  Machine Status</h2>

    <div class="machine-grid">

        <?php

            $sql = "SELECT * FROM machines";

            $result = $conn->query($sql);

            while($row = $result->fetch_assoc()) {

                $status_class = ($row['status'] == 'free') ? 'free' : 'busy';

                echo "<div class='machine ' . $status_class . '>";

                echo "<h3>" . $row['machine_name'] . "</h3>";

                if($row['status'] == 'free') {

                    echo "<p class='free'>  Free</p>";

                    echo "<p>Time: --</p>";

                } else {

                    echo "<p class='busy'>  Busy</p>";

                    echo "<p>Time: " . $row['time_remaining'] . " min left</p>";

                    echo "<p>Mode: " . $row['current_mode'] . "</p>";

                }

                echo "</div>";

            }

        ?>

    </div>

```

```
</div>
```

```
</body>
```

```
</html>
```

get_service_details.php

```
<?php
```

```
include 'db.php'
```

```
if(isset($_GET['service_id'])) {
```

```
    $service_id = $_GET['service_id'];
```

```
    $sql = "SELECT * FROM services WHERE service_id = $service_id";
```

```
    $result = $conn->query($sql);
```

```
    $row = $result->fetch_assoc();
```

```
    $response = array(
```

```
        'duration' => $row['duration_minutes'],
```

```
        'price' => $row['price']
```

```
    );
```

```
    echo json_encode($response);
```

```
}
```

```
?>
```

DATABASE DESIGN:

The screenshot shows the phpMyAdmin interface for the 'laundry_db' database. The left sidebar displays the database structure, including 'information_schema', 'laundry_db', 'mysql', 'performance_schema', 'phpmyadmin', and 'test'. The main panel shows the 'Structure' tab for 'laundry_db'. It lists four tables: 'bookings', 'customers', 'machines', and 'services'. Each table has a row of icons for actions like Browse, Structure, Search, Insert, Empty, and Drop. Below the table list, there is a 'Create new table' section with fields for 'Table name' and 'Number of columns' (set to 4), and a 'Create' button.

Table	Action	Rows	Type	Collation	Size	Overhead
bookings	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	64.0 K	18
customers	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	16.0 K	18
machines	Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	16.0 K	18
services	Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	16.0 K	18
4 tables	Sum	14	InnoDB	utf8mb4_general_ci	112.0 K	0 B

The screenshot shows the phpMyAdmin interface for the 'bookings' table in the 'laundry_db' database. The left sidebar is the same as the previous screenshot. The main panel shows the 'Browse' tab for 'bookings'. It displays a message: 'Showing rows 0 - 1 (2 total, Query took 0.0021 seconds)'. Below this is a SQL query: 'SELECT * FROM `bookings`'. There are options for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. Below these are controls for 'Show all', 'Number of rows' (set to 25), 'Filter rows' (Search this table), and 'Sort by key' (None). The table data is shown with columns: 'booking_id', 'customer_id', 'machine_id', 'service_id', 'cloth_type', 'quantity', 'total_cost', 'booking_date', 'status', and 'completion_time'. There are two rows of data. Below the table are options for 'Check all', 'With selected', 'Edit', 'Copy', 'Delete', and 'Export'. At the bottom, there is a 'Query results operations' section with buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'. There is also a 'Bookmark this SQL query' section with a 'Label' field and a checkbox 'Let every user access this bookmark'.

booking_id	customer_id	machine_id	service_id	cloth_type	quantity	total_cost	booking_date	status	completion_time
7	5	1	1	Bedding	1	80.00	2026-07-30 23:20:40	completed	NULL
8	4	1	4	Other	1	120.00	2026-07-31 00:35:24	pending	NULL

phpMyAdmin

Server: 127.0.0.1 - Database: laundry_db - Table: customers

Showing rows 0 - 1 (2 total, Query took 0.0006 seconds.)

SELECT * FROM `customers`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	customer_id	name	phone	email	address	created_at
☐	4	Shesha	9344515743	2024239027@student.annauniv.edu	qadweeldiv	2026-07-30 23:19:10
☐	5	gokul	9899888999	abc@gmail.com	drainage	2026-07-30 23:20:06

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console

phpMyAdmin

Server: 127.0.0.1 - Database: laundry_db - Table: machines

Showing rows 0 - 5 (6 total, Query took 0.0004 seconds.)

SELECT * FROM `machines`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	machine_id	machine_name	status	current_mode	time_remaining	updated_at
☐	1	Machine 1	busy	Washing + Drying	45	2026-07-31 00:35:24
☐	2	Machine 2	free	NULL	0	2026-07-30 23:10:40
☐	3	Machine 3	free	NULL	0	2026-07-30 23:10:38
☐	4	Machine 4	free	NULL	0	2026-07-30 23:10:33
☐	5	Machine 5	free	NULL	0	2026-07-30 21:47:21
☐	6	Machine 6	free	NULL	0	2026-07-30 21:47:21

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console

USER INTEREACE:

Laundry Management System

Washing Machine Status Dashboard

Menu

Dashboard

Machine Status

Book Machine

Customers

History

Machine Status Overview

Machine No.	Status	Time Remaining	Mode
Machine 1	Busy	45 min	Washing + Drying
Machine 2	Free	--	--
Machine 3	Free	--	--
Machine 4	Free	--	--
Machine 5	Free	--	--
Machine 6	Free	--	--

Summary

Total Machines: 6

Free Machines: 5

Busy Machines: 1

Laundry Management System

Washing Machine Status Dashboard

Menu

Dashboard

Machine Status

Book Machine

Customers

History

Machine Status

Machine 1

Busy

Time: 45 min left

Mode: Washing + Drying

Machine 2

Free

Time: --

Machine 3

Free

Time: --

Machine 4

Free

Time: --

Machine 5

Free

Time: --

Machine 6

Free

Time: --

Laundry Management System

Washing Machine Status Dashboard

Menu

Dashboard

Machine Status

Book Machine

Customers

History

Customer Name:

Enter customer name

Type customer name to check if they exist

Select Machine:

-- Select Machine --

Select Mode:

-- Select Mode --

Cloth Type:

Bedding

Quantity:

1

Service Details

Duration: --

Cost per item: --

Book Machine

Clear

Laundry Management System

Washing Machine Status Dashboard

Menu

Dashboard

Machine Status

Book Machine

Customers

History

Customer Management

Add New Customer

Full Name

Phone Number

Email

Address

Add Customer

Customer List

ID	Name	Phone	Email	Address	Action
5	gokul	9899888999	abc@gmail.com	drainage	Delete
4	Shesha	9344515743	2024239027@student.annauniv.edu	qadweefdv	Delete

Laundry Management System																																							
Washing Machine Status Dashboard																																							
Menu Dashboard Machine Status Book Machine Customers History																																							
Booking History <table> <tr> <th>Booking ID</th><th>Customer</th><th>Machine</th><th>Mode</th><th>Cloth Type</th><th>Quantity</th><th>Total Cost</th><th>Status</th><th>Date</th><th>Action</th></tr> <tr> <td>#8</td><td>Shesha</td><td>Machine 1</td><td>Washing + Drying</td><td>Other</td><td>1</td><td>₹120.00</td><td>PENDING</td><td>2026-07-31 00:35:24</td><td>▶ Start Processing</td></tr> <tr> <td>#7</td><td>gokul</td><td>Machine 1</td><td>Washing - Bedding</td><td>Bedding</td><td>1</td><td>₹80.00</td><td>COMPLETED</td><td>2026-07-30 23:20:40</td><td>✔ Done</td></tr> </table>										Booking ID	Customer	Machine	Mode	Cloth Type	Quantity	Total Cost	Status	Date	Action	#8	Shesha	Machine 1	Washing + Drying	Other	1	₹120.00	PENDING	2026-07-31 00:35:24	▶ Start Processing	#7	gokul	Machine 1	Washing - Bedding	Bedding	1	₹80.00	COMPLETED	2026-07-30 23:20:40	✔ Done
Booking ID	Customer	Machine	Mode	Cloth Type	Quantity	Total Cost	Status	Date	Action																														
#8	Shesha	Machine 1	Washing + Drying	Other	1	₹120.00	PENDING	2026-07-31 00:35:24	▶ Start Processing																														
#7	gokul	Machine 1	Washing - Bedding	Bedding	1	₹80.00	COMPLETED	2026-07-30 23:20:40	✔ Done																														

ADVANTAGES OF XAMPP:

Easy to Use

Provides simple control panel so that it is easy to control server components.

Local Development

Web applications can be developed and tested on a computer without deployment

Supports PHP and Databases

It provides PHP and MariaDB/MySQL, making it suitable for database-driven web applications...

Includes phpMyAdmin

The graphical interface simplifies database creation and management.

DISADVANTAGES OF XAMPP:

Not Ideal for Production

XAMPP is mainly designed for **development and testing**, not as a production server.

Security Risks

Default configurations may not provide the security required for a public-facing server.

Requires Manual Configuration

Some projects may require changes to PHP, Apache, or database configuration files.

Uses System Resources

Running Apache, database services, and other components simultaneously consumes computer memory and processing resources

.

CONCLUSION:

The Laundry Management System project successfully provides a simple and organized platform for laundry shop owners to manage customers, track machine status, handle bookings, and process billing efficiently. The integration of HTML, CSS, PHP, and MySQL with the XAMPP environment demonstrates how web and database technologies can be combined to develop a practical laundry management application. The system makes laundry operations more convenient by bringing customer records, machine tracking, order booking, and billing features into one website. Overall, the project achieves its main objective of providing a user-friendly and efficient solution for basic laundry management and can be further enhanced with features such as customer login, online payments, SMS notifications, pickup and delivery services, QR code tracking, and mobile application support.

